

1. Given the curve $\mathcal{C}$ described by the vector function $\vec{r}(t) = (2 \cos^2 t - t, 2t - \sin t, \sin(2t) + 2)$ with $-\pi/4 \leq t \leq \pi$.
 - (a) Find the unit tangent vector $\vec{T}$ at the point $(2, 0, 2)$.
 - (b) Find the curvature κ at the point in (a).
 - (c) Knowing that $\vec{B}(t) = \frac{\vec{r}'(t) \times \vec{r}''(t)}{|\vec{r}'(t) \times \vec{r}''(t)|}$, find the principal unit normal vector $\vec{N}$ to the curve $\mathcal{C}$ at the point in (a).
 - (d) Find the cartesian equation of the rectifying plane to the curve $\mathcal{C}$ at the point in (a).
2. Let the variable force $\vec{f}(t) = (\cos t, 0, 1)$ move an object of mass $m = 3$ kg along a curve $\mathcal{C}$ from an initial position $\vec{r}(0) = (1, 2, 3)$ with an initial velocity $\vec{r}'(0) = (-1, -3, 0)$. Find position $\vec{r}(t)$ of the object at any instant t . (Hint: Newton's second law of motion states that $\vec{f}(t) = m\vec{r}''(t)$).
3. Two particles travel along the space curves described by the vector functions $\vec{r}_1(t) = (\cos t, \sin t, 0)$ and $\vec{r}_2(t) = (0, \cos t, \sin t)$. Do the two particles collide? Do their paths intersect? If so, find the point(s) of intersection.
4. Sketch the domain of each of the following functions:
 - (a) $f_1(x, y) = \frac{2}{\ln(1 - x^2 - y^2)}$;
 - (b) $f_2(x, y) = \sqrt{y^2 - x^4}$;
 - (c) $f_3(x, y) = \ln(1 - x^2) + \ln(1 - y^2)$;
 - (d) $f_4(x, y) = \ln\left(\frac{1 - x^2}{1 - y^2}\right)$.